

Ray's $B_{p,q}$ Isometric Embedding Conjecture

FA/Banach discovery run `fa_banach_001`

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1 Source Question

The source paper is Samya Kumar Ray, “On Isometric Embedding $\ell_p^m \rightarrow S_\infty^n$ and Unique operator space structure”, arXiv:1911.00241. For $1 \leq p, q < \infty$, Ray defines

$$B_{p,q} = \{(z_1, z_2) \in \mathbb{C}^2 : |z_1|^p + |z_2|^q < 1\}$$

and writes $\|\cdot\|_{B_{p,q}}$ for the norm whose open unit ball is $B_{p,q}$. In Remark 4.2, after proving non-embedding for $2 < p, q < \infty$ and giving a triangular positive example, Ray conjectures that

$$(\mathbb{C}^2, \|\cdot\|_{B_{p,q}}) \hookrightarrow S_\infty \quad \text{isometrically}$$

if and only if

$$(p, q) = (2, 2), \quad (p, q) = (1, 2), \quad \text{or} \quad (p, q) = (2, 1).$$

Note that we always have for $z_2 \neq 0$, $\|(z_1, z_2)\|_p > |z_1|$. Therefore, from equation (4.4), we obtain

$$(4.5) \quad \|z_1 T_3 + z_2 S_3\|_{\ell_2 \rightarrow \ell_2} = (|z_1|^p + |z_2|^p)^{\frac{1}{p}}, \text{ for all, } z_2 \neq 0.$$

But by continuity of norm, we have from equation (4.5)

$$\|z_1 T_3 + z_2 S_3\|_{\ell_2 \rightarrow \ell_2} = (|z_1|^p + |z_2|^p)^{\frac{1}{p}}, \text{ for all, } z_1, z_2 \in \mathbb{C}.$$

Therefore, we obtain a new isometric embedding $(z_1, z_2) \mapsto z_1 T_3 + z_2 S_3$ from ℓ_p^2 to $B(\ell_2)$ where T_3 is again a positive semi-definite diagonal matrix but now there are $n - 1$ number of 1's in the diagonal. Now, we can argue as in the proof of Theorem 1.1 to obtain a contradiction. This completes the proof of the theorem. $\square$

Remark 4.2. The proof of Theorem 1.2 indicates that Theorem 1.2 holds true for any two dimensional complex Banach space $(\mathbb{C}^2, \|\cdot\|)$ with the following properties.

- (1) For all $(z_1, z_2) \in \mathbb{C}^2$, $\|(z_1, z_2)\| = \|(\overline{z_1}, \overline{z_2})\|$.
- (2) For all $0 < c \leq 1$, the operator $(z_1, z_2) \mapsto (z_1, cz_2)$ is not a contraction from $(\mathbb{C}^2, \|\cdot\|)$ to ℓ_2^2 .
- (3) $\{(z_1, z_2) \in \mathbb{C}^2 : \|(z_1, z_2)\| > |z_1|\}$ is dense in $\mathbb{C}^2$.
- (4) $\|(1, 0)\| = \|(0, 1)\| = 1$.
- (5) $\|(1, z)\| \geq 1$ and $\|(w, 1)\| \geq 1$ for all $z, w \in \mathbb{C}$.

Note that for all $2 < p, q < \infty$, the Banach spaces $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ satisfies above properties since for any $2 < p, q < \infty$, we can find $2 < p_1, p_2 < \infty$ such that $\|\cdot\|_{\ell_{p_1}} \leq \|\cdot\|_{B_{p,q}} \leq \|\cdot\|_{\ell_{p_2}}$. Hence, there is no isometric embedding of $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ into the set of all compact operators for $2 < p, q < \infty$. However, $(\mathbb{C}^2, \|\cdot\|_{B_{1,2}})$ embeds isometrically into S_∞^2 via the embedding

$$(z_1, z_2) \mapsto z_1 I_2 + z_2 e_{12}.$$

We conjecture that $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ embeds isometrically into S_∞ if and only if $(p, q) = (2, 2), (1, 2)$ or $(2, 1)$.

Proof of Theorem 1.3. If S_1^m embeds isometrically into S_p , for some $m \geq 2$, we can observe ℓ_1^2 embeds into S_p isometrically. Let $(z_1, z_2) \mapsto z_1 T + z_2 S$ be an isometry from ℓ_1^2 into S_p . Clearly, we have $T \parallel S$. Therefore, by Theorem 2.3, T and S are linearly dependent, which is absurd. This completes the proof of the theorem. $\square$

From now on, we shall always assume that Ω is a Reinhardt domain in $\mathbb{C}^2$ which is a unit ball with respect to some norm.

Proof of Theorem 1.4. We shall first show that $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ does not embed isometrically into S_∞ for $(p, q) = (2, 2)$.

2 The Result

Theorem 1. *Let $1 \leq p, q < \infty$. The two-dimensional complex Banach space $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ embeds linearly isometrically into S_∞ if and only if*

$$(p, q) \in \{(2, 2), (1, 2), (2, 1)\}.$$

3 Positive Cases

For $(p, q) = (2, 2)$, the map

$$(z_1, z_2) \mapsto \begin{pmatrix} z_1 & z_2 \\ 0 & 0 \end{pmatrix}$$

has operator norm $(|z_1|^2 + |z_2|^2)^{1/2}$.

For the mixed cases, use the elementary identity

$$\left\| \begin{pmatrix} b & a \\ 0 & b \end{pmatrix} \right\| \leq 1 \iff |a| + |b|^2 \leq 1.$$

Indeed, by rotation invariance assume $a, b \geq 0$. The largest eigenvalue of

$$\begin{pmatrix} b & a \\ 0 & b \end{pmatrix}^* \begin{pmatrix} b & a \\ 0 & b \end{pmatrix}$$

is 1 exactly when $a + b^2 = 1$, as follows by checking the determinant of $A^*A - I$, and monotonicity gives the inequality case. Hence

$$(z_1, z_2) \mapsto z_1 I_2 + z_2 e_{12}$$

is an isometry for $B_{2,1}$, while

$$(z_1, z_2) \mapsto z_2 I_2 + z_1 e_{12}$$

is an isometry for $B_{1,2}$. These matrices are finite-rank compact operators, so the range lies in S_∞ .

4 Two Obstructions

We use two negative ingredients. The first is Ray's Theorem 1.1: complex ℓ_1^2 does not embed linearly isometrically into S_∞ . This excludes $B_{1,1}$.

The second is the already-packaged partial result

$$\text{if } \max\{p, q\} > 2, \quad (\mathbb{C}^2, \|\cdot\|_{B_{p,q}}) \not\hookrightarrow S_\infty.$$

For completeness, recall the short reason. Ray's Remark 4.2 gives a criterion reducing non-embedding to the fact that for every $0 < c \leq 1$, the map $(z_1, z_2) \mapsto (z_1, cz_2)$ is not a contraction into ℓ_2^2 . If $q > 2$, choose $y > 0$ small and

$$x = (1 - y^q)^{1/p}.$$

Then (x, y) is on the $B_{p,q}$ -unit sphere and

$$x^2 + c^2 y^2 = (1 - y^q)^{2/p} + c^2 y^2 > 1$$

for all sufficiently small y , since y^2 dominates y^q . Thus the criterion applies. The case $p > 2$ follows by swapping coordinates.

It remains only to exclude the fractional region where one exponent lies strictly between 1 and 2.

5 Axis-Curve Lemma

Lemma 1. *Let $T, S \in S_\infty$ and $\|T\| = 1$. For $t \geq 0$ near 0, the function*

$$g(t) = \|T + tS\|^2$$

has the following property: if $g(t) > 1$ for all sufficiently small positive t , then

$$g(t) - 1 = ct^m + o(t^m)$$

for some integer $m \geq 1$ and some $c > 0$.

Proof. Set

$$A(t) = (T + tS)^*(T + tS) = T^*T + t(T^*S + S^*T) + t^2 S^*S.$$

This is a self-adjoint compact analytic family. Since T is compact and $\|T\| = 1$, the eigenvalue 1 of $A(0) = T^*T$ has finite multiplicity and is isolated from the rest of the spectrum. Let Γ be a small circle enclosing 1 and no other point of $\sigma(A(0))$. For small $|t|$, the Riesz projection

$$P(t) = \frac{1}{2\pi i} \int_{\Gamma} (z - A(t))^{-1} dz$$

is finite-rank and analytic in t , and the spectral values of $A(t)$ inside Γ are precisely the eigenvalues of a finite-dimensional self-adjoint analytic family. By Rellich's theorem, these eigenvalues can be written as finitely many real analytic functions

$$\lambda_1(t), \dots, \lambda_n(t), \quad \lambda_j(0) = 1.$$

The rest of the spectrum of $A(t)$ stays below $1 - \delta$ for some $\delta > 0$. Hence, for small $t \geq 0$,

$$g(t) = \max_{1 \leq j \leq n} \lambda_j(t).$$

Each $\lambda_j(t) - 1$ has an ordinary Taylor expansion with integer powers. Since $g(t) > 1$ for small positive t , at least one branch has a first nonzero Taylor coefficient that is positive on the positive side. Taking the smallest such leading order among the dominating branches gives the claimed integer m and coefficient $c > 0$. $\square$

Lemma 2. *If $1 < q < 2$, then $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ does not embed linearly isometrically into S_∞ .*

Proof. Suppose that $\Phi(z_1, z_2) = z_1T + z_2S$ is a linear isometry into S_∞ . Then $\|T\| = 1$ and, for $t \geq 0$,

$$\|T + tS\| = \|(1, t)\|_{B_{p,q}}.$$

Let $\lambda(t) = \|(1, t)\|_{B_{p,q}}$. By definition,

$$\lambda(t)^{-p} + t^q \lambda(t)^{-q} = 1.$$

Writing $\lambda(t) = 1 + u(t)$ and using $(1 + u)^{-p} = 1 - pu + O(u^2)$, we get

$$u(t) = \frac{1}{p}t^q + o(t^q).$$

Thus

$$\|T + tS\|^2 - 1 = \lambda(t)^2 - 1 = \frac{2}{p}t^q + o(t^q).$$

This contradicts the axis-curve lemma, because $q \in (1, 2)$ is not an integer. Therefore no isometric embedding exists. $\square$

Lemma 3. *If $1 < p < 2$, then $(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$ does not embed linearly isometrically into S_∞ .*

Proof. Apply the previous lemma after swapping the two coordinates, or repeat the same argument along the second coordinate axis. Explicitly, $\|(t, 1)\|_{B_{p,q}} = 1 + (1/q)t^p + o(t^p)$, which is incompatible with the integer-power conclusion of the axis-curve lemma when $1 < p < 2$. $\square$

6 Conclusion

If $\max\{p, q\} > 2$, non-embedding follows from Ray's descent criterion as recalled above. If $1 < p < 2$ or $1 < q < 2$, non-embedding follows from the axis-curve lemma. The remaining possibilities are

$$p, q \in \{1, 2\}.$$

Among these four pairs, $(1, 1)$ is excluded by Ray's complex ℓ_1^2 theorem, and $(2, 2), (1, 2), (2, 1)$ have the explicit finite-rank embeddings above. This proves the theorem.

7 Verification and Novelty Check

The script `code/check_axis_and_embeddings.py` numerically checks the axis asymptotics and the triangular positive embedding. These checks are not used as proof.

Bounded novelty check on June 13, 2026: exact web searches included “ $B_{p,q}$ ”, “ S_∞ ”, “embeds isometrically”, “Ray”, “conjecture”, and “unique operator space structure” in the relevant combinations. These searches surfaced Ray’s original arXiv page but no later resolution of the conjecture. The local run registry had only the previous $\max\{p, q\} > 2$ partial packet.

Human review should focus on the use of Rellich’s theorem in the axis-curve lemma and on the asymptotic comparison $\|(1, t)\|_{B_{p,q}} = 1 + (1/p)t^q + o(t^q)$.

Human Review: Ray's $B_{p,q}$ embedding conjecture

Original automatic proof: `solution_packet.pdf`

Checked date: 13 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears to give a correct answer to Ray's conjecture on when

$$(\mathbb{C}^2, \|\cdot\|_{B_{p,q}})$$

embeds linearly isometrically into S_∞ . The positive cases $(2,2)$, $(1,2)$, and $(2,1)$ are handled by explicit 2×2 matrix embeddings, hence by finite-rank embeddings into S_∞ .

The negative cases use Ray's existing results to exclude $(1,1)$ and the region $\max\{p,q\} > 2$. The main new part is the exclusion of the remaining fractional region, where one exponent lies strictly between 1 and 2.

Axis-Curve Lemma

The key point is the axis-curve lemma. If an isometry existed and

$$\Phi(z_1, z_2) = z_1 T + z_2 S$$

with $T, S \in S_\infty$, then along the first coordinate axis one would have

$$\|T + tS\| = \|(1, t)\|_{B_{p,q}}.$$

Since T is compact and $\|T\| = 1$, the norm-one spectral cluster of

$$(T + tS)^*(T + tS)$$

near $t = 0$ is finite-dimensional and isolated. Analytic perturbation theory then says that the relevant eigenvalues are finitely many real analytic functions of t . Consequently, if

$$\|T + tS\|^2 > 1$$

for small positive t , then

$$\|T + tS\|^2 - 1 = ct^m + o(t^m)$$

for some integer $m \geq 1$ and some $c > 0$.

Thus the compact-operator norm curve cannot have a first nonzero term with a non-integer exponent. This is exactly what rules out the fractional exponents.

Axis Asymptotic

For the $B_{p,q}$ -norm, put

$$\lambda(t) = \|(1, t)\|_{B_{p,q}}.$$

By definition,

$$\lambda(t)^{-p} + t^q \lambda(t)^{-q} = 1.$$

Since $\lambda(t) \rightarrow 1$, write

$$\lambda(t) = 1 + u(t), \quad u(t) \rightarrow 0.$$

Then

$$(1 + u)^{-p} + t^q (1 + u)^{-q} = 1.$$

Equivalently,

$$1 - (1 + u)^{-p} = t^q (1 + u)^{-q}.$$

The left-hand side satisfies

$$1 - (1 + u)^{-p} = pu + O(u^2),$$

while $(1 + u)^{-q}$ remains bounded above and below by positive constants. Therefore the equation first gives

$$u = O(t^q).$$

Using this in the expansion,

$$(1 + u)^{-p} = 1 - pu + O(u^2), \quad (1 + u)^{-q} = 1 + O(u),$$

we get

$$-pu + t^q + O(u^2) + O(t^q u) = 0.$$

Since $u = O(t^q)$, both error terms are $o(t^q)$, and hence

$$u(t) = \frac{1}{p} t^q + o(t^q).$$

It follows that

$$\|(1, t)\|_{B_{p,q}}^2 - 1 = \lambda(t)^2 - 1 = \frac{2}{p} t^q + o(t^q).$$

If $1 < q < 2$, this contradicts the integer-power conclusion of the axis-curve lemma. The same argument along the other coordinate axis, using $(t, 1)$, excludes $1 < p < 2$.

Literature and Follow-Up

The conjecture is present in the journal version of Ray's paper, on printed page 445 of the BLMS article. A preliminary literature check found later related work on Schatten-class embeddability, in particular work of Chattopadhyay–Hong–Pal–Pradhan–Ray, but did not find a later resolution of the full off-diagonal $B_{p,q}$ conjecture.

The result seems interesting. I suggest reaching out to Ray to ask whether the conjecture has since been resolved elsewhere, whether the argument is known to him, and how he would assess the significance of the proof.

Review Outcome

Archive this packet as an apparently correct full solution to Ray's $B_{p,q}$ isometric-embedding conjecture, pending author feedback and a more thorough final literature check. The proof should make explicit the $u = O(t^q)$ step in the axis asymptotic, since this is currently implicit.